IGCSE Mathematics (0580)

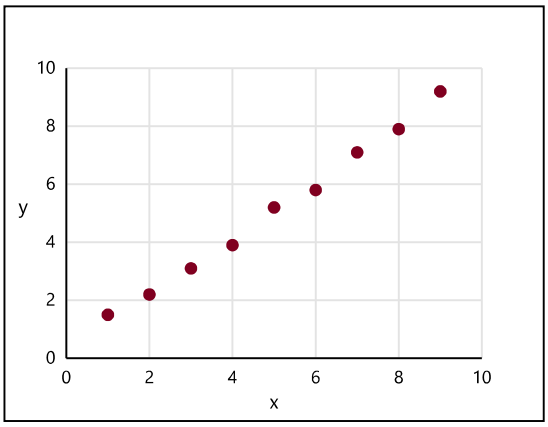
Scatter Diagrams & Managing Money · 40 Questions

Click an answer. Wrong choices turn red – keep trying until you find the green one.

Part A: Scatter Diagrams (Questions 1–10)

Question 1

Which description best fits the correlation shown in this scatter diagram?



A. No correlation

B. Strong negative correlation

C. Weak positive correlation

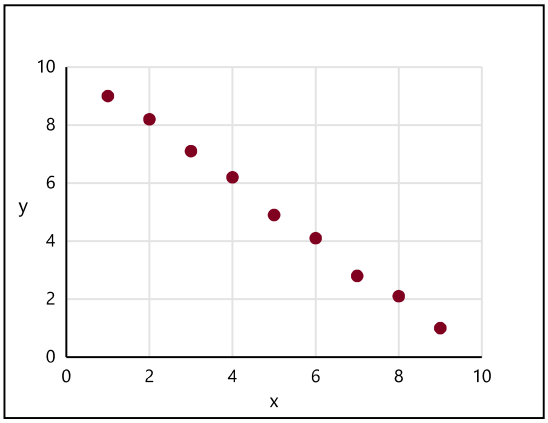
D. **Strong positive correlation ✓**

Step-by-step solution

1. The points rise from bottom-left to top-right, so the correlation is positive.
2. They lie very close to a straight line, so it is strong.
3. Answer: strong positive correlation.

Question 2

Describe the correlation shown in this scatter diagram.



A. **Strong negative correlation ✓**

B. No correlation

C. Weak negative correlation

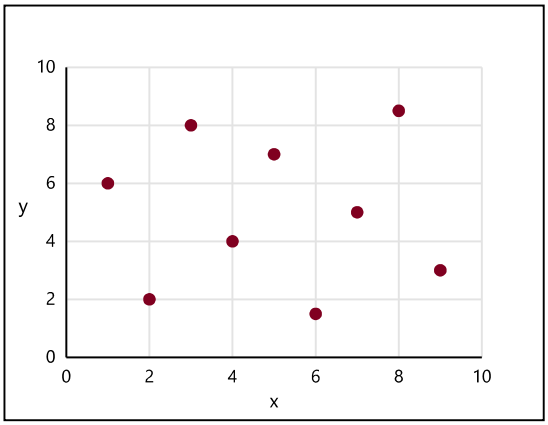
D. Strong positive correlation

Step-by-step solution

1. The points fall from top-left to bottom-right, so it is negative.
2. They lie close to a straight line, so it is strong.
3. Answer: strong negative correlation.

Question 3

What type of correlation does this scatter diagram show?



A. Weak positive correlation

B. **No correlation ✓**

C. Strong positive correlation

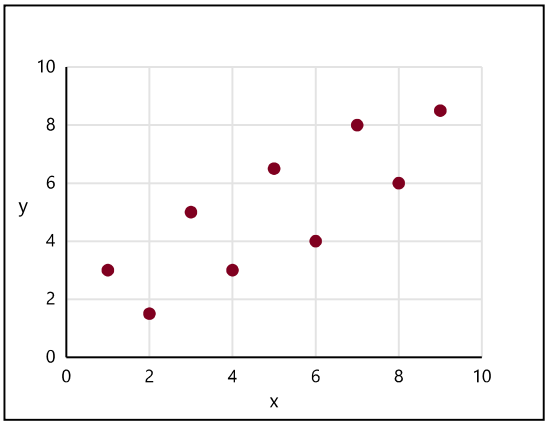
D. Strong negative correlation

Step-by-step solution

1. The points are scattered randomly with no upward or downward pattern.
2. So there is no relationship between the two variables.
3. Answer: no correlation.

Question 4

Describe the correlation shown in this scatter diagram.



A. Weak negative correlation

B. Strong positive correlation

C. **Weak positive correlation ✓**

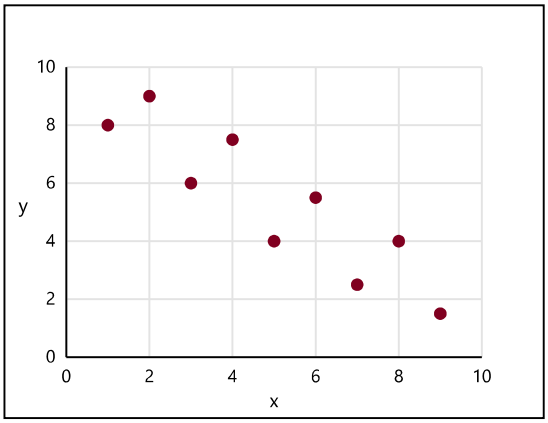
D. No correlation

Step-by-step solution

1. The general trend is upward, so the correlation is positive.
2. The points are widely spread around any line, so it is weak.
3. Answer: weak positive correlation.

Question 5

Describe the correlation shown in this scatter diagram.



A. No correlation

B. Weak positive correlation

C. Strong negative correlation

D. **Weak negative correlation ✓**

Step-by-step solution

1. The general trend is downward, so the correlation is negative.
2. The points are widely spread, so it is weak.
3. Answer: weak negative correlation.

Question 6

A shop records the daily temperature and the number of hot drinks sold. As the temperature rises, the number of hot drinks sold falls. What type of correlation is this?

A. **Negative correlation ✓**

B. Zero gradient

C. No correlation

D. Positive correlation

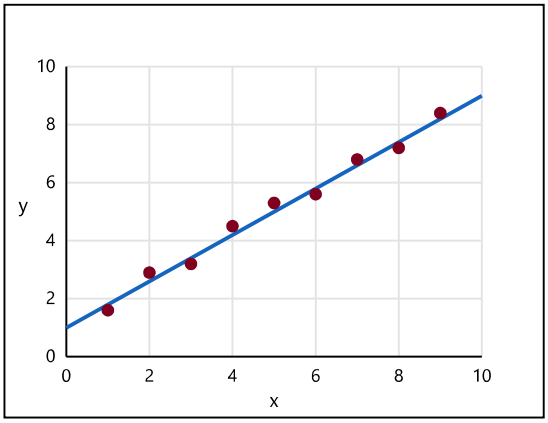
Step-by-step solution

1. As one variable increases, the other decreases.
2. This is the definition of negative correlation.

3. Answer: negative correlation.

Question 7

The scatter diagram shows a line of best fit passing through (0, 1) and (10, 9). What is its equation?



A. y = 1.25x + 1

B. **y = 0.8x + 1 ✓**

C. y = x + 1

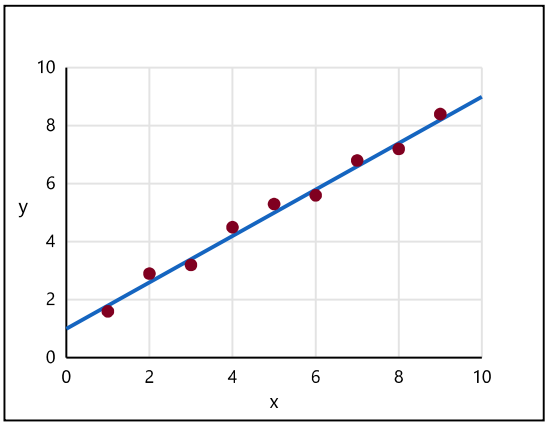
D. y = 0.8x + 9

Step-by-step solution

1. The y-intercept is where the line crosses the y-axis: c = 1.
2. Gradient = (9 − 1) ÷ (10 − 0) = 8 ÷ 10 = 0.8.
3. Equation: y = mx + c = 0.8x + 1.

Question 8

Using the line of best fit shown, estimate the value of y when x = 6.5.



A. 7.2

B. 5.2

C. **6.2 ✓**

D. 5.8

Step-by-step solution

1. Draw a vertical line up from x = 6.5 to the line of best fit.

2. Read across to the y-axis, or use the equation $y = 0.8x + 1$.
3. $y = 0.8 \times 6.5 + 1 = 5.2 + 1 = 6.2$.

Question 9

A line of best fit is drawn using data with x between 1 and 9. A student uses it to predict y when x = 40. What is the best comment?

A. The prediction is impossible with a line of best fit

B. The prediction is exact

C. The prediction is very reliable

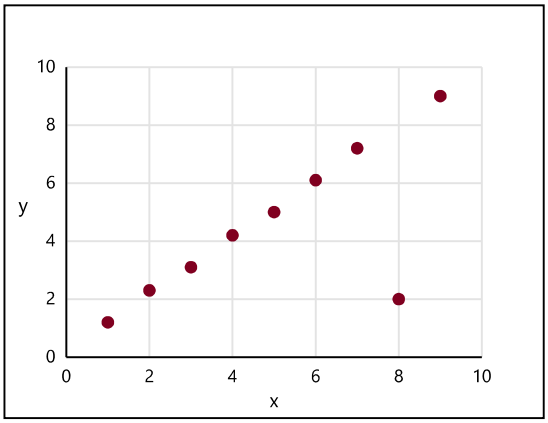
D. **The prediction is unreliable because 40 is far outside the data range ✓**

Step-by-step solution

1. Estimating outside the range of the data is called extrapolation.
2. The relationship may not continue beyond the data we measured.
3. So the prediction is unreliable.

Question 10

In this scatter diagram the point (8, 2) does not follow the trend of the other points. What is such a point called?



A. **An outlier ✓**

B. A line of best fit

C. The intercept

D. The mean point

Step-by-step solution

1. Most points follow a rising pattern.
2. (8, 2) lies far away from that pattern.
3. A point that does not fit the general trend is an outlier.

Part B: Managing Money (Questions 11–40)

Question 11

Mia earns $9.20 per hour. She works 35 hours in a week. How much does she earn?

A. $312

B. **$322 ✓**

C. $342

D. $332

Step-by-step solution

1. Earnings = hourly rate × hours.
2. 9.20 × 35 = 322.
3. She earns $322.

Question 12

Tom is paid $12 per hour for a 40-hour week. Overtime is paid at time-and-a-half. He works 6 hours overtime. What is his total pay?

A. $600

B. $552

C. **$588 ✓**

D. $624

Step-by-step solution

1. Normal pay = 12 × 40 = $480.
2. Overtime rate = 12 × 1.5 = $18 per hour.
3. Overtime pay = 18 × 6 = $108.
4. Total = 480 + 108 = $588.

Question 13

An annual salary is $36 000. What is the monthly pay?

A. $360

B. $3200

C. $2600

D. **$3000 ✓**

Step-by-step solution

1. There are 12 months in a year.
2. 36 000 ÷ 12 = 3000.
3. Monthly pay = $3000.

Question 14

Anna earns $41 600 per year. How much does she earn per week?

A. **$800 ✓**

B. $1000

C. $900

D. $700

Step-by-step solution

1. There are 52 weeks in a year.
2. 41 600 ÷ 52 = 800.
3. Weekly pay = $800.

Question 15

A salesman earns a basic salary of $1200 a month plus 4% commission on his sales. In one month he sells goods worth $15 000. What does he earn in total?

A. $1860

B. **$1800** ✓

C. $2400

D. $1260

Step-by-step solution

1. Commission = 4% of 15 000 = 0.04 × 15 000 = $600.
2. Total = basic + commission.
3. 1200 + 600 = $1800.

Question 16

Ravi's gross monthly pay is $2400. Tax of 15% and pension of 5% are deducted. What is his net pay?

A. $2160

B. $1800

C. **$1920** ✓

D. $1680

Step-by-step solution

1. Total deductions = 15% + 5% = 20%.
2. Net pay is 100% − 20% = 80% of gross pay.
3. 0.8 × 2400 = $1920.

Question 17

In a country, the first $10 000 of income is tax-free and the rest is taxed at 20%. How much tax is paid on an income of $34 000?

A. $8000

B. $3400

C. $6800

D. **$4800** ✓

Step-by-step solution

1. Taxable income = 34 000 − 10 000 = 24 000.

2. Tax = 20% of 24 000 = 0.2 × 24 000.
3. Tax = $4800.

Question 18

Find the simple interest on $2000 invested at 4% per year for 3 years.

A. **$240 ✓**

B. $250

C. $2240

D. $80

Step-by-step solution

1. Simple interest I = P × R × T ÷ 100.
2. I = 2000 × 4 × 3 ÷ 100 = 240.
3. The interest is $240.

Question 19

$5000 earns $600 simple interest in 4 years. What is the annual rate of interest?

A. 2.4%

B. **3% ✓**

C. 4%

D. 12%

Step-by-step solution

1. I = PRT ÷ 100, so R = 100I ÷ (PT).
2. R = 100 × 600 ÷ (5000 × 4) = 60 000 ÷ 20 000.
3. R = 3% per year.

Question 20

$1500 is invested at 5% simple interest per year and earns $225 interest. For how many years was it invested?

A. 4 years

B. 2 years

C. **3 years ✓**

D. 5 years

Step-by-step solution

1. Interest per year = 5% of 1500 = $75.
2. Number of years = 225 ÷ 75.
3. T = 3 years.

Question 21

Simple interest of $360 is earned in 2 years at 6% per year. What was the amount invested?

A. $4320

B. $2160

C. $6000

D. **$3000 ✓**

Step-by-step solution

1. I = PRT ÷ 100, so P = 100I ÷ (RT).
2. P = 100 × 360 ÷ (6 × 2) = 36 000 ÷ 12.
3. P = $3000.

Question 22

$4000 is invested at 5% compound interest per year. What is the total amount after 2 years?

A. **$4410 ✓**

B. $4420

C. $4200

D. $4400

Step-by-step solution

1. Multiplier = 1 + 5/100 = 1.05.
2. Amount = 4000×1.05^2.
3. $1.05^2 = 1.1025$, so amount = $4410.

Question 23

$8000 is invested at 3% compound interest per year for 3 years. How much interest is earned?

A. $720.00

B. **$741.82 ✓**

C. $760.00

D. $8741.82

Step-by-step solution

1. Amount = $8000 \times 1.03^3 = 8000 \times 1.092727 = \$8741.82$.
2. Interest = amount − original investment.
3. 8741.82 − 8000 = $741.82.

Question 24

$1000 is invested for 2 years at 10% per year. How much more interest does compound interest give than simple interest?

A. $20

B. $5

C. **$10 ✓**

D. $100

Step-by-step solution

1. Simple: 1000 × 10% × 2 = $200 interest.
2. Compound: 1000 × 1.1^2 = 1210, interest = $210.
3. Difference = 210 − 200 = $10.

Question 25

$2500 is invested at 4% compound interest per year for 5 years. Find the final amount, to the nearest cent.

A. $2600.00

B. $3125.00

C. $3000.00

D. **$3041.63 ✓**

Step-by-step solution

1. Multiplier = 1.04.
2. Amount = 2500 × 1.04^5.
3. 1.04^5 = 1.2166529..., so amount = $3041.63.

Question 26

$6000 grows to $6615 in 2 years with compound interest. What is the annual rate?

A. **5% ✓**

B. 6.15%

C. 2.5%

D. 10%

Step-by-step solution

1. Overall multiplier = 6615 ÷ 6000 = 1.1025.
2. Annual multiplier = √1.1025 = 1.05.
3. Rate = 5% per year.

Question 27

A town has a population of 20 000 growing by 2% each year. What is the population after 3 years, to the nearest whole number?

A. 21 200

B. **21 224 ✓**

C. 21 600

D. 21 000

Step-by-step solution

1. Growth multiplier = 1.02.
2. Population = 20 000 × 1.02^3.
3. 1.02^3 = 1.061208, so population ≈ 21 224.

Question 28

A population of 50 000 decreases by 4% each year. What is it after 2 years?

A. 46 208

B. 46 000

C. **46 080 ✓**

D. 46 400

Step-by-step solution

1. Decay multiplier = 1 − 0.04 = 0.96.
2. Population = 50 000 × 0.96^2.
3. 0.96^2 = 0.9216, so population = 46 080.

Question 29

A car worth $15 000 loses 20% of its value each year. What is its value after 3 years?

A. $7200

B. $9600

C. $6000

D. **$7680 ✓**

Step-by-step solution

1. Multiplier = 1 − 0.20 = 0.8.
2. Value = 15 000 × 0.8^3.
3. 0.8^3 = 0.512, so value = $7680.

Question 30

A colony of 500 bacteria increases by 10% every hour. How many bacteria are there after 4 hours, to the nearest whole number?

A. **732 ✓**

B. 740

C. 805

D. 700

Step-by-step solution

1. Growth multiplier = 1.1.
2. Number = 500 × 1.1^4.
3. 1.1^4 = 1.4641, so number = 732.05 ≈ 732.

Question 31

A trader buys a bag for $240 and sells it for $300. What is the percentage profit?

A. 20%

B. **25% ✓**

C. 125%

C. 125%

D. 60%

Step-by-step solution

1. Profit = 300 − 240 = $60.
2. Percentage profit = profit ÷ cost price × 100.
3. 60 ÷ 240 × 100 = 25%.

Question 32

A phone bought for $800 is sold for $680. What is the percentage loss?

A. 12%

B. 17.6%

C. **15% ✓**

D. 120%

Step-by-step solution

1. Loss = 800 − 680 = $120.
2. Percentage loss = loss ÷ cost price × 100.
3. 120 ÷ 800 × 100 = 15%.

Question 33

A shopkeeper sells a table for $270 and makes a 20% profit on the cost price. What was the cost price?

A. $250

B. $324

C. $216

D. **$225 ✓**

Step-by-step solution

1. Selling price = 120% of cost price = 1.2 × cost.
2. Cost = 270 ÷ 1.2.
3. Cost = $225.

Question 34

A jacket costs $80. In a sale there is a 15% discount. What is the sale price?

A. **$68 ✓**

B. $92

C. $65

D. $12

Step-by-step solution

1. Discount = 15% of 80 = $12.
2. Sale price = 80 − 12.
3. Sale price = $68 (or 0.85 × 80 = 68).

Question 35

After a 30% discount, a coat costs $63. What was the original price?

A. $81.90

B. **$90 ✓**

C. $100

D. $80

Step-by-step solution

1. The sale price is 100% − 30% = 70% of the original.
2. 0.7 × original = 63.
3. Original = 63 ÷ 0.7 = $90.

Question 36

A $200 item has 20% off, then a further 10% off the reduced price. What is the final price?

A. $146

B. $140

C. **$144 ✓**

D. $150

Step-by-step solution

1. After 20% off: 200 × 0.8 = $160.
2. After a further 10% off: 160 × 0.9.
3. Final price = $144.

Question 37

A TV costs $250 before sales tax. Sales tax is 20%. What is the price including tax?

A. $200

B. $350

C. $270

D. **$300 ✓**

Step-by-step solution

1. Tax = 20% of 250 = $50.
2. Price including tax = 250 + 50.
3. Total = $300.

Question 38

A meal costs $230 including 15% sales tax. What was the price before tax?

A. **$200 ✓**

B. $215

C. $264.50

D. $195.50

Step-by-step solution

1. Price with tax = 115% of the original = 1.15 × original.
2. Original = 230 ÷ 1.15.
3. Original = $200.

Question 39

Sam works 30 hours at $14 per hour. 12% of his gross pay is deducted. What is his net pay?

A. $370.40

B. **$369.60** ✓

C. $50.40

D. $384.00

Step-by-step solution

1. Gross pay = 30 × 14 = $420.
2. Net pay = 88% of gross = 0.88 × 420.
3. Net pay = $369.60.